`pyslfp`: theory and numerical methods

David Al-Attar
Bullard Laboratories, Department of Earth Sciences, University of Cambridge

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Abstract

This note sets out the physics and numerics behind `pyslfp`, a Python library for gravitationally self-consistent sea level change on a spherically symmetric, rotating earth, elastic or viscoelastic. It states the conventions in force throughout the code; the generalised load Love numbers, their reciprocity relations and the axial numbers that enter the rotational feedback; the spectral-element solution of the radial loading problem at each spherical harmonic degree; the linear, generalised and nonlinear forms of the sea level equation together with the iterative methods used to solve them; and the closed-form treatment of the rotational feedback. It then describes the internal-variable formulation of viscoelastic loading and its coupling to the sea level equation through an ice history, and the discrete and continuous adjoints that give sensitivity kernels with respect to the ice history, the initial sea level and the mantle viscosity. Every sign and factor below is the one the code uses.

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1 Overview

A change in the distribution of ice, water or sediment on the earth's surface deforms the solid earth, perturbs its gravitational field and alters its rotation. Each of these changes the height of the sea surface relative to the solid surface, and the water so redistributed is itself a load. The resulting pattern of relative sea level change, the *fingerprint* of the source, is the solution of the sea level equation (Farrell and Clark, 1976; Mitrovica and Milne, 2003; Kendall et al., 2005). Throughout, ξ denotes the relative sea level change and ζ the surface load, a surface mass density.

`pyslfp` computes, for a prescribed change in ice thickness (or any other surface load), the relative sea level change ξ, the radial displacement u of the solid surface, the gravitational potential perturbation ϕ at the surface, and the perturbation $\boldsymbol{\omega}$ of the angular velocity. It does so on a spherically symmetric earth whose response is entirely characterised, degree by degree, by a table of Love numbers. The library has three layers.

1. The radial earth model is held by the separate package `planetmodel`, which supplies layered models (density, elastic moduli, the radii of discontinuities), their gravity, unit systems, and radial meshes of Gauss–Lobatto–Legendre (GLL) spectral elements.

2. The module `pyslfp.love_numbers` solves the linearised, quasi-static, self-gravitating loading problem of such a model at each degree ℓ (Section 4), and returns a table of generalised Love numbers (Section 3). A default table for PREM without its ocean layer, for degrees $0 \le \ell \le 4096$, is distributed with the library.

3. The sea level solvers in `pyslfp.core` work entirely in the spherical harmonic domain on the surface: given the Love numbers they solve the sea level equation by fixed-point iteration, with the rotational feedback eliminated exactly within each iteration (Sections 5 and 6). Two thin drivers sit over this shared engine: `pyslfp.elastic`, for which a fingerprint is a single equilibrium solve, and `pyslfp.viscoelastic`, whose time stepping drives the same engine with per-step effective Love numbers (Section 7).

The class `EarthModel` binds a Love number table to a set of non-dimensional scales and spherical harmonic conventions; it adopts the body (solid surface radius a, surface gravity g and gravitational constant G) recorded with the table, so that the sea level equation and its adjoint use exactly the body for which the Love numbers were computed. A further layer, `pyslfp.linear_operators`, wraps the linear solver as a linear operator between Hilbert spaces, with its adjoint, for use in inverse problems.

2 Conventions and non-dimensionalisation

2.1 Spherical harmonics

Fields on the surface are expanded in real spherical harmonics $Y_{\ell m}$ that are orthonormal on the unit sphere,

$$\int_{\mathbb{S}^2} Y_{\ell m}\, Y_{\ell' m'}\, \mathrm{d}\Omega = \delta_{\ell\ell'}\delta_{mm'}, \tag{2.1}$$

without the Condon–Shortley phase (in `pyshtools`, normalisation `'ortho'` and `csphase=1`). Orders $m \geq 0$ carry the cosine harmonics $Y^c_{\ell m} \propto P_{\ell m}(\cos\theta)\cos m\varphi$ and $m < 0$ the sine harmonics; where the distinction matters we write $Y^c_{\ell m}$ and $Y^s_{\ell m}$. The ones needed explicitly are

$$Y_{00} = \frac{1}{\sqrt{4\pi}}, \qquad Y_{20} = \sqrt{\frac{5}{4\pi}}\,\frac{3\cos^2\theta - 1}{2}, \qquad Y^{c,s}_{21} = \sqrt{\frac{15}{4\pi}}\,\sin\theta\cos\theta\,\{\cos\varphi, \sin\varphi\}. \tag{2.2}$$

A field f on the surface has coefficients $f_{\ell m}$ with $f = \sum_{\ell m} f_{\ell m} Y_{\ell m}$, the series being truncated at a degree L (`lmax`). The surface itself is the sphere of radius b, the mean radius of the solid surface; once a Love number table is attached, $b = a$, the radius of the body for which the table was computed. Surface integrals are therefore

$$\int_{\partial M} f\,\mathrm{d}S = b^2 \int_{\mathbb{S}^2} f\,\mathrm{d}\Omega = \sqrt{4\pi}\, b^2 f_{00}, \tag{2.3}$$

and are evaluated in grid space with the exact quadrature weights of the Driscoll–Healy or Gauss–Legendre grid in use, which agrees with (2.3) for band-limited f and is exact also for products such as $C\xi$ that are not band-limited to L.

2.2 Signs

The gravitational potential is the physical potential, $\Phi = -G\int \rho/|\mathbf{x} - \mathbf{x}'|\,\mathrm{d}V'$, so that ϕ is negative near added mass and gravity is $\mathbf{g} = -\nabla\Phi$, with $g = \partial_r\Phi > 0$ in the unperturbed state. The centrifugal potential of rotation with angular velocity $\boldsymbol{\Omega}$ is $-\frac{1}{2}|\boldsymbol{\Omega}\times\mathbf{x}|^2$ with the same sign convention. Radial displacement is positive outward.

2.3 Scales

The class `EarthModelParameters` holds every physical constant in a non-dimensional form fixed by three base scales: a length L, a density P and a time T. These define a mass scale $\mathsf{M} = \mathsf{P}\mathsf{L}^3$, and a quantity with dimensions $\mathrm{M}^\alpha\mathrm{L}^\beta\mathrm{T}^\gamma$ is stored as its SI value divided by $\mathsf{M}^\alpha\mathsf{L}^\beta\mathsf{T}^\gamma$. The same three scales define a `planetmodel` unit system, so that Love numbers and radial models in

one set of units can be converted to those of the sea level solver by the same rule. The library computes in one frame, `planetmodel`'s `Scales.geophysical` at the Earth's mean radius,

$$\mathsf{L} = R_\oplus, \qquad \mathsf{P} = \bar{\rho}, \qquad \mathsf{T} = \frac{1}{\sqrt{G\bar{\rho}}}, \tag{2.4}$$

with $R_\oplus = 6371\,\mathrm{km}$ the mean radius and $\bar{\rho}$ the Earth's mean density, so that $\mathsf{T} \approx 1648\,\mathrm{s}$. In these units the radius, the mean density and the gravitational constant are one, and the surface gravity $g_0 = 9.825652323\,\mathrm{m\,s^{-2}}$ is close to $4\pi/3$. A radial model authored in other units is converted into this frame before its mesh is built, and the viscoelastic coupling refuses a model in any other. The time scale is gravitational although viscoelastic relaxation is slow: time enters the evolution only through the ratios $\Delta t/\tau$ and the conversion of dates from kiloyears, so relaxation times of order 10^7 in these units cost nothing, whereas a relaxation-based time scale would put factors of order 10^{13} into the gravitational terms of every stiffness matrix. The remaining constants are the water and ice densities $\rho_w = 1000$ and $\rho_i = 917\,\mathrm{kg\,m^{-3}}$, the rotation rate $\Omega = 7.2722 \times 10^{-5}\,\mathrm{rad\,s^{-1}}$, and the equatorial and polar moments of inertia $A = 8.0096 \times 10^{37}$ and $C = 8.0359 \times 10^{37}\,\mathrm{kg\,m^2}$. The radius b, surface gravity g and G are then replaced by those recorded with the Love number table, where the table records them.

From these the parameters derive five rotational factors, used in Section 6:

$$R_\perp = \sqrt{\frac{4\pi}{15}}\,\Omega b^2, \qquad R_z = \sqrt{\frac{16\pi}{45}}\,\Omega b^2, \qquad R_0 = -\sqrt{\frac{16\pi}{9}}\,\Omega b^2, \tag{2.5}$$

$$I_\perp = \sqrt{\frac{5}{12\pi}}\,\frac{\Omega b^3}{G(C-A)}, \qquad I_z = \sqrt{\frac{5}{9\pi}}\,\frac{\Omega b^3}{GC}. \tag{2.6}$$

In the code these are `rotation_factor`, `axial_rotation_factor`, `uniform_rotation_factor`, `inertia_factor` and `axial_inertia_factor`. The R factors map angular velocity to centrifugal potential coefficients; the I factors map potential coefficients to angular velocity through MacCullagh's formula.

3 Generalised Love numbers

3.1 Load channels

On a spherically symmetric earth a forcing of degree ℓ and order m produces a response of the same degree and order, with amplitudes independent of m. The displacement and potential are written

$$\mathbf{u} = U(r)\,Y_{\ell m}\,\hat{\mathbf{r}} + V(r)\,\nabla_1 Y_{\ell m}, \qquad \phi = \phi(r)\,Y_{\ell m}, \tag{3.1}$$

where ∇_1 is the surface gradient on the unit sphere (Dahlen and Tromp, 1998), so that

$$\int_{\mathbb{S}^2} |\nabla_1 Y_{\ell m}|^2\,\mathrm{d}\Omega = k^2 \equiv \ell(\ell+1). \tag{3.2}$$

Following Al-Attar et al. (2024, eq. 62), the action of a surface load on the earth is split into two parts: a normal traction $-g\zeta\,\hat{\mathbf{r}}$ applied to the surface, and the gravitational attraction of the surface mass, a source in Poisson's equation. Each is given its own amplitude, ζ^u for the traction-like part and ζ^ϕ for the potential source, while a true surface load ζ acts through both. A third channel, absent from the paper, is a tangential traction $-g\zeta^v\nabla_1 Y_{\ell m}$, the tangential counterpart of the traction part of a load. Finally an external potential $\psi = (r/a)^\ell Y_{\ell m}$ represents a harmonic tidal or centrifugal forcing. The surface response is

$$\begin{aligned}
u_{\ell m} &= h_\ell\,\zeta_{\ell m} + h^u_\ell\,\zeta^u_{\ell m} + h^\phi_\ell\,\zeta^\phi_{\ell m} + h^v_\ell\,\zeta^v_{\ell m} + h^t_\ell\,\psi_{\ell m},\\
v_{\ell m} &= l_\ell\,\zeta_{\ell m} + l^u_\ell\,\zeta^u_{\ell m} + l^\phi_\ell\,\zeta^\phi_{\ell m} + l^v_\ell\,\zeta^v_{\ell m} + l^t_\ell\,\psi_{\ell m},\\
\phi_{\ell m} &= k_\ell\,\zeta_{\ell m} + k^u_\ell\,\zeta^u_{\ell m} + k^\phi_\ell\,\zeta^\phi_{\ell m} + k^v_\ell\,\zeta^v_{\ell m} + k^t_\ell\,\psi_{\ell m},
\end{aligned} \tag{3.3}$$

where $u_{\ell m} = U(a)$, $v_{\ell m} = V(a)$ and $\phi_{\ell m} = \phi(a)$ are the surface values, $\psi_{\ell m}$ is the amplitude of the external potential at $r = a$, and, because a true load acts through both channels,

$$h_\ell = h_\ell^u + h_\ell^\phi, \qquad l_\ell = l_\ell^u + l_\ell^\phi, \qquad k_\ell = k_\ell^u + k_\ell^\phi. \tag{3.4}$$

The degree ℓ is written as a subscript throughout to distinguish it from the tangential Love number l. The twelve numbers $(h, l, k)_\ell^{u,\phi,v,t}$ are what the code computes and stores; the load sums are derived. They are dimensional: $h^{u,\phi,v}$ and $l^{u,\phi,v}$ are lengths per unit surface density, $k^{u,\phi,v}$ potentials per unit surface density, h^t and l^t lengths per unit potential, and k^t is dimensionless.

The split into channels is not needed to solve the sea level equation forward, for which h_ℓ and k_ℓ suffice. It is needed because the adjoint of the sea level equation is itself a sea level equation in which an observation of displacement acts through the traction channel alone and an observation of potential through the source channel alone (Al-Attar et al., 2024); similarly an observation of horizontal displacement requires the tangential channel. Section 5.4 shows how these enter the solver.

3.2 Reciprocity

The weak form of the loading problem (Section 4) is symmetric. Each channel enters as a linear functional of the test functions evaluated at the surface,

$$\zeta^u : \; -ga^2\,\hat{U}(a), \qquad \zeta^\phi : \; -a^2\,\hat{\phi}(a), \qquad \zeta^v : \; -ga^2k^2\,\hat{V}(a), \tag{3.5}$$

so the matrix of surface responses to the three load channels is symmetric up to the factors of g and of the norm k^2 of $\nabla_1 Y_{\ell m}$. Writing A for the discrete system matrix and $\mathbf{e}_U, \mathbf{e}_V, \mathbf{e}_\phi$ for the surface degrees of freedom, for instance, $k_\ell^u = -ga^2\,\mathbf{e}_\phi^{\mathsf{T}}\mathsf{A}^{-1}\mathbf{e}_U$ and $h_\ell^\phi = -a^2\,\mathbf{e}_U^{\mathsf{T}}\mathsf{A}^{-1}\mathbf{e}_\phi$. This gives the three reciprocity relations

$$g\,h_\ell^\phi = k_\ell^u, \qquad h_\ell^v = k^2\,l_\ell^u, \qquad k_\ell^v = g\,k^2\,l_\ell^\phi, \tag{3.6}$$

the first being eq. (64) of Al-Attar et al. (2024). They hold to rounding in the computed tables and are checked by `LoveNumbers.reciprocity_residual`.

3.3 Conventional numbers

The conventional dimensionless load Love numbers $h'_\ell, l'_\ell, k'_\ell$ (Farrell, 1972) are defined against the potential of the surface mass layer itself. A load $\zeta_{\ell m} Y_{\ell m}$ on the sphere of radius a has exterior potential $-4\pi G a\,\zeta_{\ell m}/(2\ell+1)$ at $r = a$, so

$$h'_\ell = \frac{(2\ell+1)\,g}{4\pi G a}\,h_\ell, \qquad l'_\ell = \frac{(2\ell+1)\,g}{4\pi G a}\,l_\ell, \qquad k'_\ell = -\frac{2\ell+1}{4\pi G a}\,k_\ell - 1, \tag{3.7}$$

where k_ℓ contains the direct attraction of the load and k'_ℓ does not. The geodetic tidal numbers are $k_\ell^T = k_\ell^t$, $h_\ell^T = -g\,h_\ell^t$ and $l_\ell^T = -g\,l_\ell^t$. In these terms the factor that maps load to sea level (Section 5) is

$$-\Big(h_\ell + \frac{k_\ell}{g}\Big) = \frac{4\pi G a}{(2\ell+1)\,g}\,(1 + k'_\ell - h'_\ell) = \frac{3}{(2\ell+1)\,\bar{\rho}}\,(1 + k'_\ell - h'_\ell), \tag{3.8}$$

with $\bar{\rho} = 3g/(4\pi G a)$ the mean density of the body.

The Green's functions for a point mass at angular distance θ follow from the addition theorem,

$$G_u(\theta) = \sum_\ell \frac{2\ell+1}{4\pi a^2}\,h_\ell\,P_\ell(\cos\theta), \qquad G_\phi(\theta) = \sum_\ell \frac{2\ell+1}{4\pi a^2}\,k_\ell\,P_\ell(\cos\theta), \tag{3.9}$$

and are summed in the code with the taper $\exp(-10\ell^2/L^2)$.

3.4 Degrees zero and one

At $\ell = 1$ the problem is posed in the frame of the centre of mass of the earth and its load. The degree-one exterior potential of a body is proportional to the offset of its centre of mass from the origin, so in this frame it vanishes and $\phi(a) = 0$ for every forcing. The potential source channel then does no work, and

$$h_1^\phi = l_1^\phi = k_1^\phi = 0, \qquad k_1 = 0, \qquad k_1' = -1. \tag{3.10}$$

At $\ell = 0$ there is no tangential displacement, and a uniform external potential is a gauge, so the tidal numbers vanish. The load numbers do not: conservation of mass fixes the exterior potential of the loaded body,

$$k_0 = -4\pi G a, \qquad k_0' = 0, \tag{3.11}$$

while h_0 is the radial compression of the earth under the weight of a uniform layer.

3.5 Axial numbers

Five further scalars describe the degree-zero response more completely; they are needed for the component of the rotational feedback along the rotation axis (Section 6). A change ω_z in spin rate has a centrifugal potential whose spherical mean is proportional to r^2, not constant, and so is not harmonic; and any uniform radial deformation changes the trace of the inertia tensor. Neither effect is contained in the surface response to a load or to a harmonic tide. We define

h_c, k_c: the surface displacement and potential at degree zero per unit centrifugal potential $(r/a)^2 Y_{00}$;

m_u, m_ϕ, m_c: the inertia moments

$$m = \int_M \rho\, \mathbf{u} \cdot \mathbf{x}\, \mathrm{d}V = \sqrt{4\pi} \int_0^a \rho\, U\, r^3\, \mathrm{d}r \tag{3.12}$$

of the degree-zero responses to unit forcing in the traction channel, the source channel and the centrifugal potential respectively.

The moment (3.12) is one quarter of the change in the trace of the inertia tensor produced by the displacement, since $\delta \int \rho |\mathbf{x}|^2 \mathrm{d}V = 2 \int \rho\, \mathbf{u} \cdot \mathbf{x}\, \mathrm{d}V$ for a mass-conserving displacement and $\operatorname{tr} \mathbf{I} = 2 \int \rho |\mathbf{x}|^2 \mathrm{d}V$. The shell theorem gives $k_c = 0$: a spherically symmetric redistribution of mass inside the body leaves its exterior potential unchanged.

The centrifugal forcing enters the weak form as the functional $-\int \rho r^2\, \partial_r \psi\, \hat{U}\, \mathrm{d}r$ with $\psi = (r/a)^2$ (Section 4.2), that is $-(2/a^2) \int \rho r^3 \hat{U}\, \mathrm{d}r = -(2/(\sqrt{4\pi} a^2))\, \hat{m}$, where $\hat{m}$ is the moment (3.12) of the test function. The same symmetry argument as for (3.6) then gives

$$m_u = \tfrac{1}{2}\sqrt{4\pi}\, g\, a^4\, h_c, \qquad m_\phi = \tfrac{1}{2}\sqrt{4\pi}\, a^4\, k_c, \tag{3.13}$$

checked by `axial_reciprocity_residual`; in particular $m_\phi = 0$, a surface shell attracting nothing inside it. The self-moment $m_c = -\frac{1}{2}\sqrt{4\pi} a^2\, \mathbf{b}_c^{\mathsf{T}} \mathsf{A}^{-1} \mathbf{b}_c$ is negative when A is positive definite: a positive centrifugal potential $(r/a)^2$ pulls inward. A table that lacks the axial numbers (as read from a file without them) has them undefined, and the sea level solver then sets the deformation terms of the axial feedback to zero.

3.6 The table

`LoveNumbers` holds the twelve numbers by degree, the five axial numbers, and the body (a, g, G), in the units of a given scale system. It is computed by `LoveNumbers.from_model` from any `planetmodel` model whose surface is solid, or read from a plain-text file in SI units whose header records the columns, the body and the axial numbers. A model frozen at a frequency ω, whose moduli are complex, yields complex numbers for frequency-domain viscoelastic work; the sea level solver accepts only real tables.

4 The radial problem at degree ℓ

4.1 The bilinear form

The Love numbers are surface values of the solution of the linearised, quasi-static, self-gravitating elastic loading problem on a spherically symmetric, non-rotating, transversely isotropic earth, reduced by the expansion (3.1). The weak form is eq. (D76) of Al-Attar and Tromp (2014), whose isotropic elastic block we write for a transversely isotropic medium with Love's moduli A, C, F, L, N. Define, for the radial scalars of (3.1),

$$X = r\partial_r U, \qquad Y = 2U - k^2 V, \qquad S = U + r\partial_r V - V, \tag{4.1}$$

and similarly $\hat{X}, \hat{Y}, \hat{S}$ for test functions $\hat{w} = (\hat{U}, \hat{V}, \hat{\phi})$. In solid regions I_S the integrand is

$$\begin{aligned}\mathcal{E}_S(w,\hat{w}) = {} & CX\hat{X} + (A-N)\,Y\hat{Y} + F\,(X\hat{Y} + Y\hat{X}) + k^2 L\,S\hat{S} + k^2(k^2-2)\,N\,V\hat{V} \\ & + 4\rho\,(\pi G\rho r - g)\,r\,U\hat{U} + k^2\rho g r\,(U\hat{V} + V\hat{U}) \\ & + \rho r^2\,(\hat{U}\,\partial_r\phi + U\,\partial_r\hat{\phi}) + k^2\rho r\,(V\hat{\phi} + \phi\hat{V}) + \frac{1}{4\pi G}(r^2\,\partial_r\phi\,\partial_r\hat{\phi} + k^2\phi\hat{\phi}),\end{aligned} \tag{4.2}$$

with $\rho(r)$ the density and $g(r)$ the gravity of the reference model. Under $A = C = \kappa + \frac{4}{3}\mu$, $F = \kappa - \frac{2}{3}\mu$ and $L = N = \mu$ the elastic block becomes

$$\kappa\,(X+Y)(\hat{X}+\hat{Y}) + \tfrac{4}{3}\mu\,(X - \tfrac{1}{2}Y)(\hat{X} - \tfrac{1}{2}\hat{Y}) + k^2\mu\,S\hat{S} + k^2(k^2-2)\,\mu\,V\hat{V}, \tag{4.3}$$

which is the form of Al-Attar and Tromp (2014): $X + Y$ is r times the dilatation, and $X - \frac{1}{2}Y$, S and V are the three deviatoric strain scalars of the spheroidal field.

In fluid regions I_F, above degree zero, the quasi-static problem does not determine the displacement: equilibrium requires only that the perturbed level surfaces of pressure, density and potential coincide. An equipotential is displaced radially by $-\phi/g$, so the Eulerian density perturbation is $\rho_1 = (\partial_r\rho/g)\,\phi$ and the fluid enters through the potential alone, with integrand

$$\mathcal{E}_F(\phi,\hat{\phi}) = \frac{1}{4\pi G}(r^2\,\partial_r\phi\,\partial_r\hat{\phi} + k^2\phi\hat{\phi}) + \frac{r^2\,\partial_r\rho}{g}\,\phi\hat{\phi}. \tag{4.4}$$

The stratification term is taken as zero where $g = 0$. At each fluid–solid boundary $r = d$ the integration by parts leaves the point term

$$\pm\,\rho_f\,d^2\big(g\,U\hat{U} + \phi\,\hat{U} + U\hat{\phi}\big)\big|_{r=d}, \tag{4.5}$$

where ρ_f is the fluid density at the boundary and the sign is positive when the fluid lies below (as at the core–mantle boundary) and negative when it lies above (as at the inner-core boundary), U being the value on the solid side.

The gravitational field outside the domain is eliminated by Dirichlet-to-Neumann maps. Outside $r = a$ the potential is $\phi(a)(a/r)^{\ell+1}$, whose energy $\frac{1}{4\pi G}\int_a^\infty (r^2\phi'^2 + k^2\phi^2)\mathrm{d}r$ is $\frac{(\ell+1)a}{4\pi G}\phi(a)^2$. When the domain is truncated below at $r_0 > 0$ (Section 4.5), the potential below is continued as the regular harmonic $\phi(r_0)(r/r_0)^\ell$, with energy $\frac{\ell r_0}{4\pi G}\phi(r_0)^2$. The full form is therefore

$$\begin{aligned}\mathcal{A}_\ell(w,\hat{w}) = {} & \int_{I_S}\mathcal{E}_S\,\mathrm{d}r + \int_{I_F}\mathcal{E}_F\,\mathrm{d}r + \sum_d (\pm)\rho_f d^2(gU\hat{U} + \phi\hat{U} + U\hat{\phi}) \\ & + \frac{(\ell+1)a}{4\pi G}\phi(a)\hat{\phi}(a) + \frac{\ell r_0}{4\pi G}\phi(r_0)\hat{\phi}(r_0),\end{aligned} \tag{4.6}$$

symmetric in its two arguments. The problem for a forcing $\mathcal{F}$ is to find w with $\mathcal{A}_\ell(w,\hat{w}) = \mathcal{F}(\hat{w})$ for all admissible $\hat{w}$.

4.2 Forcings

The three load channels are the functionals (3.5) per unit amplitude; a true surface load is the sum of the first two, $-a^2\,[g\hat{U}(a)+\hat{\phi}(a)]$, which is eq. (D79) of Al-Attar and Tromp (2014). An external potential $\psi(r)Y_{\ell m}$ is moved to the right-hand side term by term: wherever ϕ couples to the displacement or to the fluid density perturbation in (4.2), (4.4) and (4.5) it is replaced by the total potential $\phi+\psi$, the self-gravitation term $(4\pi G)^{-1}|\nabla\phi|^2$ being unchanged. This gives

$$\mathcal{F}_\psi(\hat{w}) = -\int_{I_S} \rho(r^2\,\partial_r\psi\,\hat{U} + k^2 r\,\psi\,\hat{V})\mathrm{d}r - \int_{I_F} \frac{r^2\,\partial_r\rho}{g}\,\psi\,\hat{\phi}\,\mathrm{d}r - \sum_d (\pm)\rho_f d^2\,\psi(d)\,\hat{U}(d). \qquad (4.7)$$

The harmonic tide is $\psi=(r/a)^\ell$. The only non-harmonic case used is $\psi=(r/a)^2$ at $\ell=0$, the spherical mean of the centrifugal potential of a change in spin rate, which gives h_c, k_c and m_c. The constant potential, $\psi=1$, gives zero force.

4.3 Degrees zero and one

At $\ell=1$ the surface potential degree of freedom is removed, imposing $\phi(a)=0$ and with it the centre-of-mass frame; the exterior Dirichlet-to-Neumann term goes with it. No forcing then acts through the potential source channel, whose functional is $-a^2\hat{\phi}(a)$.

At $\ell=0$ there is no tangential degree of freedom, and the fluid regions are treated as elastic media with $\mu=0$ rather than through (4.4): for a purely radial deformation the fluid's displacement is determined by its compressibility, and (4.3) with $k^2=0$ reduces to $\kappa(r\partial_r U+2U)^2$, the energy of radial compression. The condition $U(0)=0$ is imposed at the centre.

4.4 Discretisation

The radial interval is divided into spectral elements whose edges include every discontinuity of the model. On each element the unknowns are Lagrange polynomials on n GLL nodes (`ngll`, by default five, so quartic elements), and all integrals are evaluated by the GLL quadrature on the same nodes. Radial derivatives are $\partial_r = J^{-1}\mathsf{D}$, with D the GLL differentiation matrix on the reference element and J the element Jacobian. Solid nodes carry (U,V,ϕ), fluid nodes ϕ alone, with U and V at a fluid–solid boundary belonging to the solid side; the degrees of freedom are numbered node by node, so the global matrix is banded with half-bandwidth $k_d=3(n-1)+2$. Moduli, density and gravity are sampled at the nodes element by element, so that a node shared by two layers carries both one-sided values.

The matrix is real symmetric for real moduli and complex symmetric otherwise, and is stored as an upper band. It is first scaled symmetrically by its diagonal, $\tilde{\mathsf{A}}=\mathsf{D}_A^{-1/2}\mathsf{A}\mathsf{D}_A^{-1/2}$, which equalises the very different units of the displacement and potential unknowns. A real system is then factorised by banded Cholesky; the gravitational terms and the fluid stratification term mean that positive definiteness is not guaranteed, and the code falls back to banded LU when the factorisation fails. A complex system is solved by banded LU. For each degree one factorisation serves all forcings: the three load channels and the tide, and at $\ell=0$ the centrifugal potential.

4.5 Sub-mesh and graded mesh

At high degree the solution decays rapidly with depth, roughly as $(r/a)^{\ell+1}$, and is below a fraction ε (`eps`, by default 10^{-8}) of its surface value deeper than $D_\ell = a\,(1-\varepsilon^{1/(\ell+1)})$. Each degree $\ell>0$ is therefore solved on the sub-mesh of elements above this depth, closed below by the interior Dirichlet-to-Neumann term of (4.6); the displacement and potential below are set to zero.

A uniform mesh adequate for degree ℓ has elements no wider than $0.1a/(\ell+1)$. At depth d below the surface the highest degree whose solution reaches that depth satisfies $\ell+1=$

$\ln\varepsilon/\ln(1-d/a)$, so an element there need be no narrower than $0.1a\,\ln(1-d/a)/\ln\varepsilon$. The graded mesh used by default for a table to degree L has element width

$$w(r) = \max\Big(\frac{0.1\,a}{L+1},\ \frac{0.1\,a\,\ln(1-d/a)}{\ln\varepsilon}\Big), \qquad d = a - r, \tag{4.8}$$

about $d/184$ at depth for $\varepsilon = 10^{-8}$. Elements are laid down from the top of each layer of the model downwards; the last span above a discontinuity is divided into one or two equal elements, none wider than w at its top. Every degree thereby sees at least the resolution the uniform rule would give it throughout its own sub-mesh, while the whole mesh has a few hundred elements rather than the $10(L+1)$ of the uniform rule at L.

5 The sea level equation

5.1 The linear equation

Let C be the ocean function of the background state, one over the oceans and zero elsewhere, $\mathcal{A}_o = \int C\,\mathrm{d}S$ the ocean area, and ζ_d a prescribed direct load, for example $\rho_i\,\Delta I$ for a change ΔI in grounded ice thickness. With the shorelines held fixed the ocean responds with a water load $\rho_w C\xi$, so the total load is

$$\zeta = \zeta_d + \rho_w\, C\,\xi. \tag{5.1}$$

The sea surface remains an equipotential of the gravitational and centrifugal potentials, so its height changes by $-(\phi+\psi)/g$ plus a constant, and the relative sea level change, the change in the height of the sea surface above the solid surface, is

$$\xi = -u - \frac{\phi+\psi}{g} + c, \tag{5.2}$$

where $u_{\ell m} = h_\ell \zeta_{\ell m}$ and $\phi_{\ell m} = k_\ell \zeta_{\ell m}$ apart from the rotational terms of Section 6, and c is a spatial constant. The constant is fixed by conservation of mass, the water added to the oceans being the mass removed from the direct load,

$$\rho_w \int C\,\xi\,\mathrm{d}S = -\int \zeta_d\,\mathrm{d}S \quad\Longrightarrow\quad c = \bar{\xi}_E - \frac{1}{\mathcal{A}_o}\int C\,\tilde{\xi}\,\mathrm{d}S, \qquad \bar{\xi}_E = -\frac{1}{\rho_w \mathcal{A}_o}\int \zeta_d\,\mathrm{d}S, \tag{5.3}$$

where $\tilde{\xi} = -u - (\phi+\psi)/g$ and $\bar{\xi}_E$ is the eustatic (barystatic) sea level change. Equations (5.1)–(5.3) are the linear sea level equation of Farrell and Clark (1976), with the rotational terms of Milne and Mitrovica (1998).

5.2 Fixed-point structure

Combining (5.1) and (5.2), the equation is a fixed point for the load,

$$\zeta = \mathcal{G}(\zeta) \equiv \zeta_d + \rho_w\, C\,\mathcal{P}[\mathcal{S}\zeta], \qquad (\mathcal{S}\zeta)_{\ell m} = -\Big(h_\ell + \frac{k_\ell}{g}\Big)\zeta_{\ell m} + (\text{rotation}), \tag{5.4}$$

where $\mathcal{P}$ adds the constant (5.3). The iteration works on the coefficients of the load. Each evaluation of $\mathcal{G}$ multiplies the load coefficients by the combined factor $-(h_\ell + k_\ell/g)$, replaces the degree-2 coefficients of orders 0 and 1 by those of Section 6, synthesises $\tilde{\xi}$ on the grid with one inverse transform, applies the mass-conservation shift using the grid quadrature, multiplies by $\rho_w C$, adds ζ_d, and returns to coefficients with one forward transform. The displacement and potential are synthesised only once, from the converged load. The starting load is the eustatic one, $\zeta^{(0)} = \zeta_d + \rho_w C\,\bar{\xi}_E$.

By (3.8) the linear part of $\mathcal{G}$ is the product of multiplication by C, with norm one on L^2, and a spherical harmonic multiplier of size $3\rho_w(1+k'_\ell - h'_\ell)/[(2\ell+1)\bar{\rho}]$, with the degree-zero part removed by the mass constraint. Since $\rho_w/\bar{\rho} \approx 0.18$ and the multiplier decays as $1/(2\ell+1)$, the plain Picard iteration contracts, most slowly at the lowest degrees.

5.3 Anderson acceleration

The iteration is accelerated by Anderson mixing (Walker and Ni, 2011). With x_k the current vector of load coefficients, $f_k = \mathcal{G}(x_k) - x_k$ its residual, and the last p differences $\Delta x_i = x_{i+1} - x_i$, $\Delta f_i = f_{i+1} - f_i$ gathered as the columns of ΔX_k and ΔF_k, the update is

$$\gamma_k = \arg\min_{\gamma} \left\| f_k - \Delta F_k \gamma \right\|_2, \qquad x_{k+1} = \mathcal{G}(x_k) - (\Delta X_k + \Delta F_k)\gamma_k. \tag{5.5}$$

The least-squares problem is solved through its normal equations $(\Delta F_k^{\mathsf{T}} \Delta F_k)\gamma_k = \Delta F_k^{\mathsf{T}} f_k$, the $p \times p$ Gram matrix being updated by one row and column per iteration from ring buffers of the differences, with a minimum-norm least-squares solve guarding against rank deficiency. The default memory is $p = 5$; $p = 0$ recovers the Picard iteration. The iteration stops when the change of the ocean sea level between iterates satisfies

$$\frac{\max |C\xi^{(k)} - C\xi^{(k-1)}|}{\max |C\xi^{(k)}|} \leq \texttt{rtol}, \tag{5.6}$$

with $\texttt{rtol} = 10^{-9}$ by default.

5.4 The generalised equation and the adjoint

The generalised sea level equation adds three forcings to the direct load: a displacement load ζ^u acting through the traction channel, a potential load ζ^ϕ acting through the source channel, and an angular momentum forcing $\mathbf{L}$ in the rotational equations. The first two contribute static terms independent of the iteration,

$$u^s_{\ell m} = h^u_\ell \zeta^u_{\ell m} + h^\phi_\ell \zeta^\phi_{\ell m}, \qquad \phi^s_{\ell m} = k^u_\ell \zeta^u_{\ell m} + k^\phi_\ell \zeta^\phi_{\ell m}, \tag{5.7}$$

which are added to u and ϕ, so that $\tilde{\xi}_{\ell m}$ gains $-(u^s_{\ell m} + \phi^s_{\ell m}/g)$; their degree-2 and degree-0 coefficients also enter the rotational feedback (Section 6). Mass conservation (5.3) involves ζ_d alone.

The reason for these channels is the adjoint. The fingerprint operator $\mathcal{F} : \zeta_d \mapsto (\xi, u, \phi, \boldsymbol{\omega})$, with the L^2 inner product on the sphere of radius b for the fields and the Euclidean one for $\boldsymbol{\omega}$, has an adjoint that is again a solution of the generalised equation (Al-Attar et al., 2024). Its adjoint applied to $(\xi^\dagger, u^\dagger, \phi^\dagger, \boldsymbol{\omega}^\dagger)$ is the relative sea level change of the generalised equation with

$$\zeta_d = \xi^\dagger, \qquad \zeta^u = -u^\dagger, \qquad \zeta^\phi = -g\,\phi^\dagger, \qquad \mathbf{L} = -g(\boldsymbol{\omega}^\dagger - \Psi^* \phi^\dagger), \tag{5.8}$$

where $\Psi : \boldsymbol{\omega} \mapsto \psi$ is the centrifugal potential map of (6.2) below and

$$\Psi^* f = b^2 (R_\perp f^c_{21},\ R_\perp f^s_{21},\ R_z f_{20} + R_0 f_{00}) \tag{5.9}$$

its adjoint. This identity is exact only when the solver uses the reciprocal numbers of a single table and the body (a, g, G) of that table, which is why `EarthModel` adopts the body from the table.

5.5 The nonlinear equation with migrating shorelines

For large changes of ice the ocean function itself changes, and the background must be carried as fields: the ice thickness I and the sea level S, the height of the sea surface above the solid surface (ocean depth where positive, minus the topography on land). The ocean function is set by flotation,

$$C = H(\rho_w S - \rho_i I), \tag{5.10}$$

with H the Heaviside function, so that ice thinner than the water depth that would float it is part of the ocean, and only grounded ice loads the earth as ice. Optionally a closed basin (the

Caspian) is excluded from C. Given an initial state (I_0, S_0, C_0), an ice thickness change ΔI, a sediment thickness change ΔH_s of density ρ_s ($2300\,\mathrm{kg\,m^{-3}}$ by default) and a dynamic sea level change ξ_{dyn}, the final ice is $I = I_0 + \Delta I$ and the solver iterates on (S_n, C_n) from (S_0, C_0). Each iteration forms the load

$$\zeta_n = \rho_i\big[(1 - C_n)I - (1 - C_0)I_0\big] + \rho_w(C_n S_n - C_0 S_0) + \rho_s \Delta H_s, \tag{5.11}$$

computes $\tilde{\xi}_n = \mathcal{S}\zeta_n$ as in (5.4), including the rotational terms but without the constant, and the provisional sea level $\hat{S} = S_0 + \tilde{\xi}_n - \Delta H_s + \xi_{\mathrm{dyn}}$, sediment raising the solid surface. The constant is a eustatic shift that conserves the total of water, grounded ice and sediment over the current ocean,

$$\Delta_n = \frac{M_n - \rho_w \int C_n \hat{S}\,\mathrm{d}S}{\rho_w \int C_n\,\mathrm{d}S}, \tag{5.12}$$

$$M_n = \rho_w \int C_0 S_0\,\mathrm{d}S - \int \rho_i\big[(1 - C_n)I - (1 - C_0)I_0\big]\mathrm{d}S - \int \rho_s \Delta H_s\,\mathrm{d}S, \tag{5.13}$$

after which

$$\xi_{n+1} = \tilde{\xi}_n + \Delta_n, \qquad S_{n+1} = \hat{S} + \Delta_n, \qquad C_{n+1} = H(\rho_w S_{n+1} - \rho_i I). \tag{5.14}$$

This is the formulation of Kendall et al. (2005). The iteration is plain Picard, stopped when $\max|\xi_{n+1} - \xi_n| \le \texttt{rtol}\,\max|\xi_n|$; the error is measured on the whole grid, not only over the ocean. The result is a new state (I, S_{n+1}) together with ξ, u, ϕ and $\boldsymbol{\omega}$, the last three synthesised from the load of the final iteration.

6 Rotational feedback

6.1 Centrifugal potential

Let the angular velocity be $\Omega\hat{\mathbf{z}} + \boldsymbol{\omega}$. To first order the centrifugal potential changes by

$$\psi = -\Omega\omega_z r^2 + \Omega z\,(\boldsymbol{\omega}\cdot\mathbf{x}) = \Omega r^2\Big[\sin\theta\cos\theta\,(\omega_x\cos\varphi + \omega_y\sin\varphi) + \omega_z\big(\cos^2\theta - \tfrac{1}{3}\big) - \tfrac{2}{3}\,\omega_z\Big]. \tag{6.1}$$

With (2.2), its coefficients on the surface $r = b$ are

$$\psi^c_{21} = R_\perp\,\omega_x, \qquad \psi^s_{21} = R_\perp\,\omega_y, \qquad \psi_{20} = R_z\,\omega_z, \qquad \psi_{00} = R_0\,\omega_z, \tag{6.2}$$

with the factors (2.5). The degree-2 parts are of the form $(r/b)^2 Y_{2m}$, harmonic, and act on the earth through the tidal Love numbers h^t_2, k^t_2. The degree-0 part is $\psi_{00}(r/b)^2 Y_{00}$, not harmonic, and acts through the axial numbers h_c, k_c and m_c. The *transverse* components ω_x, ω_y (true polar wander) therefore couple to degree 2, order 1, and the *axial* component ω_z (change in length of day) to degree 2, order 0, and degree 0.

6.2 Liouville equation and MacCullagh's formula

On the time scales of loading the Chandler wobble is not excited and the linearised Liouville equation reduces to its quasi-static balance. With $\Delta\mathbf{I}$ the perturbation of the inertia tensor and $\mathbf{h}$ a relative angular momentum, it reads

$$(C - A)\,\omega_x = \Omega\,\Delta I_{xz} + h_x, \qquad (C - A)\,\omega_y = \Omega\,\Delta I_{yz} + h_y, \qquad C\,\omega_z = -\Omega\,\Delta I_{zz} - h_z. \tag{6.3}$$

The inertia perturbation is read from the degree-2 potential by MacCullagh's formula. Expanding $-G\int\rho/|\mathbf{x} - \mathbf{x}'|$ to second order outside the mass and writing $\Delta\mathbf{I}$ with $\Delta T = \operatorname{tr}\Delta\mathbf{I}$,

$$\phi^c_{21} = \frac{3G}{b^3}\sqrt{\frac{4\pi}{15}}\,\Delta I_{xz}, \qquad \phi^s_{21} = \frac{3G}{b^3}\sqrt{\frac{4\pi}{15}}\,\Delta I_{yz}, \qquad \phi_{20} = \frac{G}{b^3}\sqrt{\frac{\pi}{5}}\,(3\Delta I_{zz} - \Delta T). \tag{6.4}$$

Eliminating $\Delta\mathbf{I}$ between (6.3) and (6.4) gives, with the factors (2.6),

$$\omega_{x,y} = I_\perp\,\phi_{21}^{c,s} + \frac{h_{x,y}}{C-A}, \qquad \omega_z = -I_z\,\phi_{20} - \frac{\Omega}{3C}\,\Delta T - \frac{h_z}{C}. \tag{6.5}$$

The transverse equations involve only the degree-2 potential. The axial equation also involves the trace ΔT, which the exterior potential does not see. The angular momentum forcing of the generalised equation enters as $\mathbf{h} = -\mathbf{L}$; that is, $\mathbf{L}$ adds $-\mathbf{L}_{x,y}/(C-A)$ to $\omega_{x,y}$ and $+L_z/C$ to ω_z.

6.3 The trace

The change in trace has two sources. A surface mass layer σ on the sphere of radius b contributes $2b^2\int\sigma\,\mathrm{d}S = 2\sqrt{4\pi}\,b^4\sigma_{00}$, and the mass-like sources are the true load and the potential load, so $\sigma_{00} = \zeta_{00} + \zeta_{00}^{\phi}$; the displacement load is a traction and carries no mass. The deformation contributes $4m$, with m the moment (3.12) of the degree-0 displacement, which by Section 3.5 is

$$m = (m_u + m_\phi)\,\zeta_{00} + m_u\,\zeta_{00}^{u} + m_\phi\,\zeta_{00}^{\phi} + m_c\,\psi_{00}. \tag{6.6}$$

Hence

$$\frac{\Omega}{3C}\,\Delta T = \frac{4\Omega}{3C}\,m + \frac{2\Omega\sqrt{4\pi}\,b^4}{3C}\,(\zeta_{00} + \zeta_{00}^{\phi}). \tag{6.7}$$

6.4 Closed-form fixed points

The potential in (6.5) contains the response to the centrifugal potential, $\phi_{2m} = \phi_{2m}^{L} + k_2^t\psi_{2m}$, where $\phi_{2m}^{L} = k_2\zeta_{2m} + \phi_{2m}^{s}$ is the potential of the loads. Likewise the moment depends on $\psi_{00} = R_0\omega_z$. Since (6.2) is linear in $\boldsymbol{\omega}$, each component is a scalar fixed point, solved in closed form:

$$\omega_{x,y} = \frac{I_\perp\,\phi_{21}^{L,c/s} - L_{x,y}/(C-A)}{1 - I_\perp\,k_2^t\,R_\perp}, \tag{6.8}$$

$$\omega_z = \frac{-I_z\,\phi_{20}^{L} - \dfrac{4\Omega}{3C}\,[(m_u+m_\phi)\zeta_{00} + m_u\zeta_{00}^{u} + m_\phi\zeta_{00}^{\phi}] - \dfrac{2\Omega\sqrt{4\pi}\,b^4}{3C}\,(\zeta_{00}+\zeta_{00}^{\phi}) + L_z/C}{1 + I_z\,k_2^t\,R_z + \dfrac{4\Omega}{3C}\,m_c\,R_0}. \tag{6.9}$$

In the axial denominator both corrections are positive ($k_2^t > 0$, and $m_c < 0$, $R_0 < 0$), so the feedback of the axial deformation on the spin rate is stabilising. In the linear equation ζ_{00} vanishes at convergence by mass conservation, so the degree-0 terms in ζ matter chiefly for the generalised equation and for loads that do not conserve mass.

Given $\boldsymbol{\omega}$, the rotational contributions to the fields are

$$u_{2m} \mathrel{+}= h_2^t\psi_{2m}, \quad \phi_{2m} \mathrel{+}= k_2^t\psi_{2m}, \quad u_{00} \mathrel{+}= h_c\psi_{00}, \quad \phi_{00} \mathrel{+}= k_c\psi_{00}, \tag{6.10}$$

for the degree-2 coefficients of orders 0 and 1 and the degree-0 coefficient, and the relative sea level coefficients become

$$\tilde\xi_{2m} = -\Big(u_{2m} + \frac{\phi_{2m}+\psi_{2m}}{g}\Big), \qquad \tilde\xi_{00} \mathrel{-}= \Big(h_c + \frac{k_c+1}{g}\Big)\psi_{00}. \tag{6.11}$$

The last is uniform and is absorbed by the mass-conservation constant; it is kept so that $\tilde\xi$ is the complete local response. The returned potential ϕ is the gravitational perturbation without ψ.

Because the rotational fixed point is solved exactly inside every evaluation of $\mathcal{G}$, the angular velocity, displacement and potential returned are all consistent with the same load, and the outer iteration has only the ocean loading to converge.

6.5 Rotational stability and the choice of moments

The equilibrium moments of inertia in (6.8) and (6.9) are specified quantities: they cannot be computed from the spherically symmetric model in which the elastic problem is solved, whose inertia tensor is isotropic (Yu et al., 2025). The transverse denominator $1 - I_\perp k_2^t R_\perp$ equals $1 - k_2^t/k_{\rm sus}$, with $k_{\rm sus} = 3G(C-A)/(\Omega^2 b^5)$ the oblateness the specified moments sustain, and it must remain positive for every tidal response the model can produce. The largest such response is the hydrostatic fluid limit $k_{\rm hyd}$, the degree-2 tidal number of the model with its rigidity set to zero, which depends only on the density profile (Kelvin's $3/(2(\ell-1))$ for a homogeneous body); the library computes it directly as the zero-rigidity solid rather than as the $s \to 0$ limit of a viscoelastic $k_2^t(s)$, which degrades numerically (Mitrovica et al., 2005). For the Earth the observed moments satisfy $k_{\rm hyd} < k_{\rm sus}$, the excess being the non-hydrostatic part of the figure, and the specified observed values are the correct default. For any other body the moments may instead be constructed from the model itself in the manner of Mitrovica et al. (2005): $C - A = (1+\delta)\, k_{\rm hyd}\Omega^2 b^5/(3G)$ with δ a prescribed non-hydrostatic fraction (0.8 per cent for the Earth), which makes the margin $\delta/(1+\delta)$ whatever the rheology. A configuration with $k_{\rm hyd} \geq k_{\rm sus}$ has a transverse feedback on the wrong side of the secular resonance and is refused by the time-domain solvers; the axial feedback is unconditionally stable, both denominator corrections in (6.9) being positive, and may be retained alone.

7 Viscoelastic loading

This section describes the formulation implemented in `pyslfp.viscoelastic`.

7.1 Rheology and internal variables

The extension treats a generalised Maxwell body with branches $k = 1, \ldots, K$ of shear modulus μ_k and relaxation time τ_k, and a long-term shear modulus μ_∞, in the internal-variable formulation of Al-Attar and Tromp (2014) (see also Crawford et al., 2017). The same formulation underlies the time-domain treatment of glacial isostatic adjustment, and of its sensitivity kernels by the adjoint method, in laterally heterogeneous earth models by Crawford et al. (2018); here the earth is spherically symmetric and each degree decouples. With d the deviatoric strain, each branch carries a trace-free symmetric internal variable m_k, and

$$\mathbf{T} = \kappa\,(\operatorname{tr}\boldsymbol{\varepsilon})\,\mathbf{1} + 2\mu_\infty d + \sum_k 2\mu_k\,(d - m_k), \qquad \dot{m}_k = \frac{d - m_k}{\tau_k}. \tag{7.1}$$

The unrelaxed shear modulus is $\mu_U = \mu_\infty + \sum_k \mu_k$, which is the elastic modulus of the material. The classical Maxwell body is one branch with $\mu_\infty = 0$ and $\tau = \eta/\mu$. The viscosity may be prescribed directly, or through the relaxation time $\tau(r)$, in which case $\eta = \mu\tau$ inherits its depth dependence from the shear modulus. Bulk relaxation is not modelled, fluid regions carry no branches, and branches require an isotropic material.

7.2 The degree-basis scalars

For spheroidal forcing the internal variable of each branch reduces, at each (ℓ, m), to three radial scalars (Al-Attar and Tromp, 2014, eqs D68–D73): M, the purely tangential component, R, the radial–radial component, and S, the radial–tangential component; the toroidal components remain zero. They are driven by the deviatoric strain scalars of (4.3),

$$e_M = V, \qquad e_R = r\partial_r U - U + \tfrac{1}{2}k^2 V = X - \tfrac{1}{2}Y, \qquad e_S = r\partial_r V - V + U = S, \tag{7.2}$$

branch by branch,

$$\dot{M}_k = \frac{e_M - M_k}{\tau_k}, \qquad \dot{R}_k = \frac{e_R - R_k}{\tau_k}, \qquad \dot{S}_k = \frac{e_S - S_k}{\tau_k}, \tag{7.3}$$

the strain scalars being evaluated at the GLL nodes by collocation. The weights with which they enter the μ-part of (4.3) are $\frac{4}{3}$, k^2 and $k^2(k^2-2)$ for R, S and M.

7.3 Quasi-static equilibrium and the memory force

At fixed internal variables the displacement and potential satisfy the elastic problem with the unrelaxed moduli, forced by the loads and by a memory force:

$$\mathcal{A}_\ell^U(w,\hat{w}) = \mathcal{F}(\hat{w}) + b_{\mathrm{mem}}(\hat{w}), \tag{7.4}$$

$$b_{\mathrm{mem}}(\hat{w}) = \sum_k \int \mu_k \Big[\tfrac{4}{3} R_k\, e_R(\hat{w}) + k^2 S_k\, e_S(\hat{w}) + k^2(k^2-2)\, M_k \hat{V}\Big]\mathrm{d}r, \tag{7.5}$$

where $\mathcal{A}_\ell^U$ is (4.6) assembled with μ_U. This is the one-sided form of the μ-part of (4.3) applied to the internal variables, and it follows from (7.1) by moving the $-2\mu_k m_k$ stress to the right-hand side. At $\ell = 0$ only the R term survives; at $\ell = 1$ the M term vanishes although M still evolves. The state of the system is the internal variables alone: the displacement, potential and, once coupled, the sea level are slaved to them through degree solves, each a banded back-substitution with the orders as right-hand-side columns.

7.4 Time stepping

With $h_k = \Delta t/\tau_k$, each scheme eliminates the new internal variable in favour of the new strain, which leaves an elastic solve with an effective shear modulus and a memory force built from known quantities.

Backward Euler: $m^{n+1} = (m^n + h\, d^{n+1})/(1+h)$, so that

$$\mu_{\mathrm{eff}} = \mu_\infty + \sum_k \frac{\mu_k}{1+h_k}, \qquad \tilde{m}_k = \frac{m_k^n}{1+h_k}. \tag{7.6}$$

Exponential trapezoid: with $e = \exp(-h)$, $\varphi_1 = (1-e)/h$, $\alpha = \varphi_1 - e$ and $\beta = 1 - \varphi_1$, the exact update for a strain linear in time over the step is $m^{n+1} = e\, m^n + \alpha\, d^n + \beta\, d^{n+1}$, so that

$$\mu_{\mathrm{eff}} = \mu_\infty + \sum_k \varphi_{1,k}\, \mu_k, \qquad \tilde{m}_k = e_k\, m_k^n + \alpha_k\, d^n; \tag{7.7}$$

φ_1 is evaluated by its Taylor series for small h.

Exponential Euler: $m^{n+1} = e\, m^n + (1-e)\, d^n$, explicit, the solve at t^n using the unrelaxed moduli and $b_{\mathrm{mem}}(m^n)$.

In each case the step's equilibrium is $\mathcal{A}_\ell^{\mathrm{eff}}(w,\hat{w}) = \mathcal{F}(\hat{w}) + b_{\mathrm{mem}}(\tilde{m})$, after which m^{n+1} follows from the update formula. An elastic region is the limit $\tau \to \infty$: $h = 0$, $e = 1$, $\alpha = \beta = 0$, $\varphi_1 = 1$, and m remains zero. Backward Euler is first order and the exponential trapezoid second order; the exponential schemes are exact for relaxation under strain prescribed piecewise constant (exponential Euler) or piecewise linear (trapezoid) in time, for any Δt. One factorisation per degree and per (scheme, Δt) serves every step.

7.5 Coupling to the sea level equation

With the effective moduli the instantaneous response of a step to a surface load is described by per-step effective Love numbers, computed by the elastic machinery of Section 4 on the effective material, together with a memory field, the surface response (U, ϕ) to $b_{\mathrm{mem}}(\tilde{m})$ under zero load. The step's sea level equation is then the generalised linear equation of Section 5 with the effective numbers in place of h_ℓ, k_ℓ and the memory field as a static term, closed by the same iteration for

the ocean and rotational feedbacks. The converged load gives d^{n+1} at the nodes and completes the update of the internal variables. In the limit $\tau_k \to \infty$ for every branch the formulation reduces to the elastic one. The same body can also be treated in the frequency domain, through the complex moduli of a model frozen at a frequency (Section 3); by the correspondence principle (Wu and Peltier, 1982) the two descriptions are related by a Laplace transform, which provides an independent check of the time stepping.

A run starts at its first time t_0 with the internal variables zero and the instantaneous response, computed with the unrelaxed numbers, to the load at t_0: the right limit, so that a load switched on at the start is stepped at full order rather than costing the trapezoid an order.

7.6 Ice histories and shorelines

An ice history supplies the thickness $I(t)$ at any date, interpolating linearly between its slices; dates are in kiloyears before present and the step grid should contain the history's breakpoints. With the shorelines frozen at those of the background state, the direct load at the nth date is the grounded thickness change $\zeta_{d,n} = \rho_i(1 - C_0)(I_n - I_0)$, and each step's sea level equation is the linear one with the ocean function C_0. With the shorelines migrating, each step solves the nonlinear equation of Section 5 from the background state, with the memory field as a ready response whose displaced water the explicit mass budget conserves, and the load of the converged state relative to the background finishes the radial step. The ocean function C_n of each step is recorded: it is all of the trajectory that the adjoint of Section 8 needs. Under migrating shorelines the step is consistent to the tolerance of the nonlinear iteration rather than eliminated exactly. In either case the transverse rotational feedback is refused when the specified moments cannot sustain the model's hydrostatic limit (Section 6).

8 Sensitivity kernels

8.1 Objective functionals and the adjoint problem

An objective functional of an evolution is a sum of measurements at observation dates t_i on the step grid,

$$J = \sum_i \Big[\int g_\xi^{(i)}\, \xi(t_i)\, \mathrm{d}S + \int g_u^{(i)}\, u(t_i)\, \mathrm{d}S + \int g_\phi^{(i)}\, \phi(t_i)\, \mathrm{d}S + \mathbf{g}_\omega^{(i)} \cdot \boldsymbol{\omega}(t_i) \Big], \tag{8.1}$$

and its kernels are the derivatives with respect to the ice thickness $I(\mathbf{x}, t)$, the initial sea level $S_0(\mathbf{x})$ and the inverse relaxation times $\tau_k^{-1}(r)$. The adjoint theory of glacial isostatic adjustment was set out by Crawford et al. (2018) and extended to rotational feedbacks by Yu et al. (2025), whose formulation is followed here. The adjoint problem has the form of the forward problem with more general forces: g_ξ forces like an ice load, g_u and g_ϕ enter as the displacement and potential channels of the generalised equation, and $\mathbf{g}_\omega$ as an angular momentum forcing, through the mapping (5.8). The adjoint ocean function is the time-reversed ocean function of the forward run rather than a dynamical variable, so the adjoint problem is linear whether or not the shorelines migrate, and the singular terms of shoreline motion cancel. Under frozen shorelines the ice history enters linearly and no forward run is needed.

The library realises the adjoint twice: as the exact transpose of the time-stepped map, and by discretising the continuous adjoint equations. A discrete adjoint is exact for the discretised functional, which in an under-resolved run is a poor stand-in for the continuum one; the two differ by the commute error of discretisation and transposition, which vanishes as the run resolves its rheology, so their disagreement is a diagnostic of the forward resolution.

8.2 The discrete adjoint of a step

Per degree, let K be the symmetric effective system of the step, D the map from a degree solution to the nodal strain scalars, W the nodal weights $\mu_k w_q w_c$ (the GLL quadrature weight times the scalar weight w_c of M, R or S), E, A and B the diagonal scheme weights (e, α and β for the trapezoid), P the surface rows giving $U(a)$ and $\phi(a)$, and $\mathbf{q}$ the row giving the degree-0 inertia moment. The memory-force assembly is $\mathsf{D}^{\mathsf{T}}\mathsf{W}$, with the same quadrature. One step, as a linear map of the state $(\mathbf{m}^n, \mathbf{d}^n)$, is

$$\begin{gathered}\tilde{\mathbf{m}} = \mathsf{E}\mathbf{m}^n + \mathsf{A}\mathbf{d}^n, \qquad \mathbf{x}_m = \mathsf{K}^{-1}\mathsf{D}^{\mathsf{T}}\mathsf{W}\tilde{\mathbf{m}}, \qquad (U_m, \phi_m) = \mathsf{P}\mathbf{x}_m, \quad J_0 = \mathbf{q}\cdot\mathbf{x}_m,\\ (\sigma, \psi) = \mathcal{L}(\zeta_{d,n+1}, (U_m, \phi_m), J_0), \qquad \mathbf{x} = \mathbf{x}_m + \mathbf{x}_\sigma\sigma^{\mathsf{T}} + \mathbf{x}_\psi\psi^{\mathsf{T}},\\ \mathbf{d}^{n+1} = \mathsf{D}\mathbf{x}, \qquad \mathbf{m}^{n+1} = \tilde{\mathbf{m}} + \mathsf{B}\mathbf{d}^{n+1},\end{gathered}$$

where $\mathcal{L}$ is the step's generalised sea level solve, returning the converged load and centrifugal coefficients, and $\mathbf{x}_\sigma$ and $\mathbf{x}_\psi$ are the unit solutions for a load and a tide, the load right-hand sides being rank one. With daggers for duals, the transpose run in reverse is

$$\begin{gathered}\mathbf{x}^\dagger = \mathsf{D}^{\mathsf{T}}(\mathsf{B}\mathbf{m}^{n+1\dagger} + \mathbf{d}^{n+1\dagger}), \qquad \sigma^\dagger = \mathbf{x}_\sigma\cdot\mathbf{x}^\dagger, \quad \psi^\dagger = \mathbf{x}_\psi\cdot\mathbf{x}^\dagger,\\ (\zeta^\dagger_{d,n+1}, (U_m, \phi_m)^\dagger, J_0^\dagger) = \mathcal{L}^{\mathsf{T}}(\sigma^\dagger, \psi^\dagger), \qquad \mathbf{x}_m^\dagger = \mathbf{x}^\dagger + \mathsf{P}^{\mathsf{T}}(U_m, \phi_m)^\dagger + \mathbf{q}\,J_0^\dagger,\\ \tilde{\mathbf{m}}^\dagger = \mathbf{m}^{n+1\dagger} + \mathsf{W}\mathsf{D}\mathsf{K}^{-1}\mathbf{x}_m^\dagger, \qquad \mathbf{m}^{n\dagger} = \mathsf{E}\tilde{\mathbf{m}}^\dagger, \qquad \mathbf{d}^{n\dagger} = \mathsf{A}\tilde{\mathbf{m}}^\dagger,\end{gathered}$$

the measurement duals at date $n+1$ joining the inputs of $\mathcal{L}^{\mathsf{T}}$. A reverse step costs one symmetric banded solve per degree, as a forward step does. Backward Euler is the case $\mathsf{E} = (1+h)^{-1}$, $\mathsf{A} = 0$, $\mathsf{B} = 1 - \mathsf{E}$, and the sweep closes with the transpose of the right-limit solve at t_0.

The transpose $\mathcal{L}^{\mathsf{T}}$ is the exact reverse mode of the engine's own code path. The generalised solve is a fixed point for the load built from a diagonal Love-number multiplication, multiplication by the ocean function, the synthesis and analysis transforms (mutual transposes up to the quadrature weights), a rank-one mass-conservation projection and the closed-form rotational scalars, each of which transposes directly, so the adjoint is a transposed fixed point run through the same Anderson mixer, returning the duals of the direct load, both response channels and the inertia moment. Its rotational block dualises the complete degree-2 displacement and potential, not only their forcing parts. Under migrating shorelines the same construction applies about the trajectory, with the recorded C_n in place of C_0.

8.3 The continuous adjoint

Between observation times the adjoint state evolves by the forward equations without load, in reversed time, so it is stepped by the forward machinery. At an observation time the measurement drives an instantaneous, unrelaxed generalised fingerprint problem through the mapping (5.8); its sea level gives the singular part of the ice kernel at that time, and the adjoint internal variables jump by

$$[\mathbf{m}^\dagger] = \frac{1}{\tau}\,\tilde{\mathbf{d}}^\dagger, \tag{8.2}$$

with $\tilde{\mathbf{d}}^\dagger$ the strain of the instantaneous adjoint solution (Yu et al., 2025, eq. 80). The smooth adjoint fields are densities per unit time, and the date quadrature splits its weight at each observation across the jump. This formulation never sees the stepper's internals, and survives a change of scheme.

8.4 The ice kernel and its views

The discrete adjoint returns, for each date, the dual λ_n of the direct load, $\delta J = \sum_n \int \lambda_n\,\delta\zeta_{d,n}\,\mathrm{d}S$. Each load depends on the thickness at its own date through $\rho_i(1 - C_n)I_n$ and on the initial

thickness through $-\rho_i(1-C_0)I_0$, so the thickness duals are $\rho_i(1-C_n)\lambda_n$ with the baseline $-\rho_i(1-C_0)\sum_n \lambda_n$ added at the first date; the baseline grounds with the initial ocean function. These satisfy the gradient identity exactly for perturbations on the date grid, but they do not converge pointwise under refinement, because the kernel is singular in time (Yu et al., 2025, eq. 81):

$$K_I(\mathbf{x},t) = \kappa(\mathbf{x},t) + \sum_i \kappa_i(\mathbf{x})\,\delta(t-t_i) + \kappa_0(\mathbf{x})\,\delta(t-t_0). \tag{8.3}$$

A load applied at an observation time produces an instantaneous elastic response, whose amplitude κ_i is the grounded instantaneous fingerprint adjoint of that measurement, independent of the date grid; κ_0 is the baseline; and the density κ converges under refinement. The quelled view of Backus (1970) removes the singularities by changing variable to the rate of thickness change, $I(t) = I(t_0) + \int \dot{I}$: the kernel with respect to $\dot{I}$ is the cumulative time integral of K_I, bounded with finite jumps at the observation times only, and pairs with histories continuous in time. Under frozen shorelines the dual of the initial thickness then vanishes identically, a uniform thickness change loading nothing.

8.5 The initial sea level

The ocean load at a date is $\rho_w(C_nS_n - C_0S_0)$ with $S_n = S_0 + \xi_n$, so the initial sea level enters each load as $\rho_w(C_n - C_0)\,\delta S_0$, and its kernel assembles from the load duals already computed,

$$K_{S_0} = \rho_w \sum_n (C_n - C_0)\,\lambda_n, \tag{8.4}$$

the discrete form of eq. (65) of Yu et al. (2025). It vanishes identically under frozen shorelines. This is the kernel for objectives on the sea level change; an objective on the absolute sea level $S_0 + \xi$ adds the sum of its own sea level duals.

8.6 Relaxation times and viscosity

With the forward trajectory recorded, the adjoints also return $\partial J/\partial \tau_k^{-1}(r)$ for each branch, the one-dimensional reduction of eq. (61) of Yu et al. (2025), whose tensor contraction becomes the weighted sum over the scalars M, R and S. The discrete variant differentiates the scheme weights in $h = \Delta t\,\tau^{-1}$: per step, the derivatives of e and α pair $\tilde{\mathbf{m}}^\dagger$ with the forward $\mathbf{m}^n$ and $\mathbf{d}^n$, and the derivative of β pairs $\mathbf{m}^{n+1\dagger}$ with $\mathbf{d}^{n+1}$ and, through the effective modulus, the transposed and forward solutions through the branch bilinear form. The continuous variant integrates the multiplier of the internal-variable dynamics, the branch-weighted adjoint state over τ^{-1}, against the forward $\mathbf{d} - \mathbf{m}$. The viscosity kernels follow from $\partial\tau^{-1}/\partial\eta = -\tau^{-2}/\mu$, so an elastic branch has no viscosity sensitivity in this parameterisation.

Recording the trajectory at every date is expensive at high degree, so it may be checkpointed and reconstructed backwards between stored states. The inverse of a trapezoid step is the same update with the step negated and the start and end strains exchanged, since $e(-h) = 1/e(h)$, $\alpha(-h) = -\beta(h)/e(h)$ and $\beta(-h) = -\alpha(h)/e(h)$; its effective weight $\varphi_1(-h) > 1$ keeps the system positive definite. Backward Euler inverts explicitly, $\mathbf{m}^n = (1+h)\mathbf{m}^{n+1} - h\,\mathbf{d}^{n+1}$. Running dissipation backwards amplifies rounding by e^h per branch and step, so the sum of h across a gap is the number of digits spent.

8.7 Decay times

A decay time is the parameter $\tau_d = 1/s$ of a curve $a\,e^{-st} + b$ fitted to a series v_j at times t_j, so its kernel follows from differentiating the fit. Following Al-Attar and Tromp (2014, Appendix C), the model times $\hat{t}_j$ join the parameters with Gaussian penalties towards the nominal ones,

to account for the timing uncertainty of sea level data, and the fit minimises

$$\sum_j \frac{\left(v_j - a\,e^{-s\hat{t}_j} - b\right)^2}{\sigma_v^2} + \sum_j \frac{(\hat{t}_j - t_j)^2}{\sigma_t^2}, \tag{8.5}$$

only the ratio of the two uncertainties mattering. The implicit function theorem applied to the stationarity conditions, with the full Hessian, gives the weights $w_j = \partial\tau_d/\partial v_j$, so that the decay time is, to first order, the multi-date measurement $\sum_j w_j v_j$ to which every kernel above applies. The datum belongs near the centre of a former ice sheet and after the melting has stopped, where rebound curves are close to exponential; elsewhere the fit degenerates towards long decay times and is refused.

9 The initial sea level

An ice history that carries no sea level leaves the initial sea level S_0 to be chosen, and the standard device is to choose it so that the run reproduces a final sea level S_f at the last date t_f, the observed present day on the Earth. The run from S_0 reaches

$$\mathcal{H}(S_0) = S_0 + \xi(t_f; S_0), \tag{9.1}$$

and the calibration solves $\mathcal{H}(S_0) = S_f$ or, more generally, minimises

$$J = \frac{1}{2}\int w\left(\mathcal{H}(S_0) - S_f\right)^2 \mathrm{d}S \tag{9.2}$$

with non-negative weights w. The linearisation is $\mathcal{H}' = I + A$, with $A = \partial\xi(t_f)/\partial S_0$ arising only from the dependence of the loads on S_0.

Under frozen shorelines the problem is linear. The shorelines are those of the starting state, held fixed, so ξ does not depend on S_0 at all, $A = 0$, and one run gives $S_0 = S_f - \xi(t_f)$ exactly wherever $w > 0$. The calibrated state keeps those shorelines as a prescribed ocean function, so that a frozen run from it reproduces S_f. Were the ocean function instead re-derived from each new S_0 by flotation, the problem would no longer be linear, and cells near the flotation limit could flip back and forth between iterations without settling. Under migrating shorelines the initial shorelines do follow from S_0 by flotation, and the loads at later dates depend on S_0 through the terms $\rho_w(C_n - C_0)S_0$ of Section 8, so that A is confined to where the shorelines move, and the calibration iterates.

The classical update (Kendall et al., 2005), with $r = \mathcal{H}(S_0) - S_f$,

$$S_0 \leftarrow S_0 - r = S_f - \xi(t_f; S_0), \tag{9.3}$$

is Newton's method for $\mathcal{H}(S_0) = S_f$ with $I + A$ replaced by the identity, contracting at the rate set by A; with $A = 0$ it is the frozen-shoreline solution. The gradient of (9.2) in the surface pairing is

$$\nabla J = (I + A^{\mathsf{T}})\, w\, r, \tag{9.4}$$

the weighted residual plus the kernel (8.4) of a sea level measurement $w\,r$ at t_f (Yu et al., 2025, eq. 87). With $A = 0$ and unit weights a unit step down it is (9.3), which is the sense in which the classical scheme approximates steepest descent (Yu et al., 2025, Section 3.9). Under migrating shorelines unit steps contract at about the rate set by $A + A^{\mathsf{T}}$, more slowly than (9.3), which inverts the Jacobian approximately where the gradient applies its transpose; the gradient is the method for objectives that (9.3) does not serve, a weighted or partial comparison with the present. Either iteration is accelerated by Anderson mixing.

The connection with Kalman filtering places the two. Let S_0 have prior mean S_b and covariance P, and let the final sea level be observed as $y = \mathcal{H}(S_0) + \varepsilon$ with noise covariance R. The

Kalman update iterated about the current estimate, which is Gauss–Newton on the posterior (Bell and Cathey, 1993), is

$$S_0 \leftarrow S_b + K\big[y - \mathcal{H}(S_0) - \mathcal{H}'(S_b - S_0)\big], \qquad K = P\,\mathcal{H}'^{\mathsf{T}}\big(\mathcal{H}'P\,\mathcal{H}'^{\mathsf{T}} + R\big)^{-1}. \tag{9.5}$$

For a perfect and complete observation, $R \to 0$, the gain is $K = \mathcal{H}'^{-1}$, the prior drops out and the iteration is Newton's method for $\mathcal{H}(S_0) = y$; with $\mathcal{H}'$ replaced by the identity the gain is $K = I$ and the iteration is (9.3). The classical scheme is an iterated Kalman update of unit gain, exact in a single step under frozen shorelines. The adjoint supplies the action of $\mathcal{H}'^{\mathsf{T}}$ that the gain needs beyond that case: the calibration is a strong-constraint variational problem for the initial state, which for linear Gaussian problems coincides with the Kalman smoother, the adjoint run being the smoother's backward sweep. An imperfect or incomplete present-day field makes the gain a blend of prior and data, $K = P(P + R)^{-1}$ when $\mathcal{H}' = I$; the weights of (9.2) are the simplest instance, the inverse of a diagonal noise covariance without a prior.

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